PRIMITIVE DATA TYPES

Int / short int / signed int / unsigned int / long int

float – for decimal no's ➔ 4 bytes ➔ $3.4*10^{-38}$ to $3.4*10^{+38}$ - %f

long float / double – 8 bytes - %lf ➔ $1.7*10^{-308}$ to $1.7*10^{+308}$

long double ➔ 10 bytes - %Lf ➔ $3.4*10^{-4932}$ to $1.1*10^{+4932}$

char ➔ 256 ASCII characters

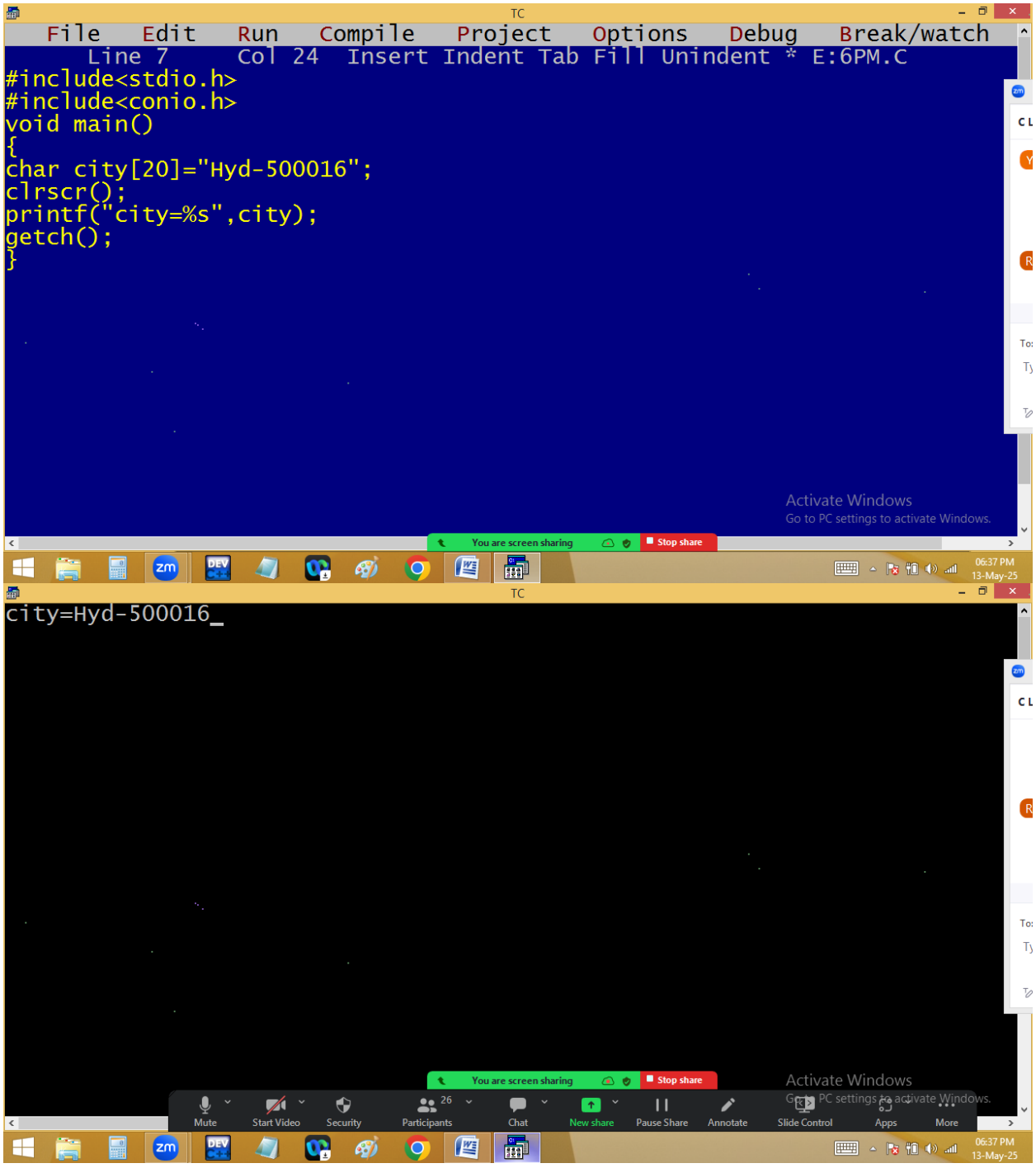
1. Signed char ➔ -128 to +127 ➔ %c ➔ 1 byte
2. Unsigned char ➔ 0 to 255 ➔ %c ➔ 1 byte

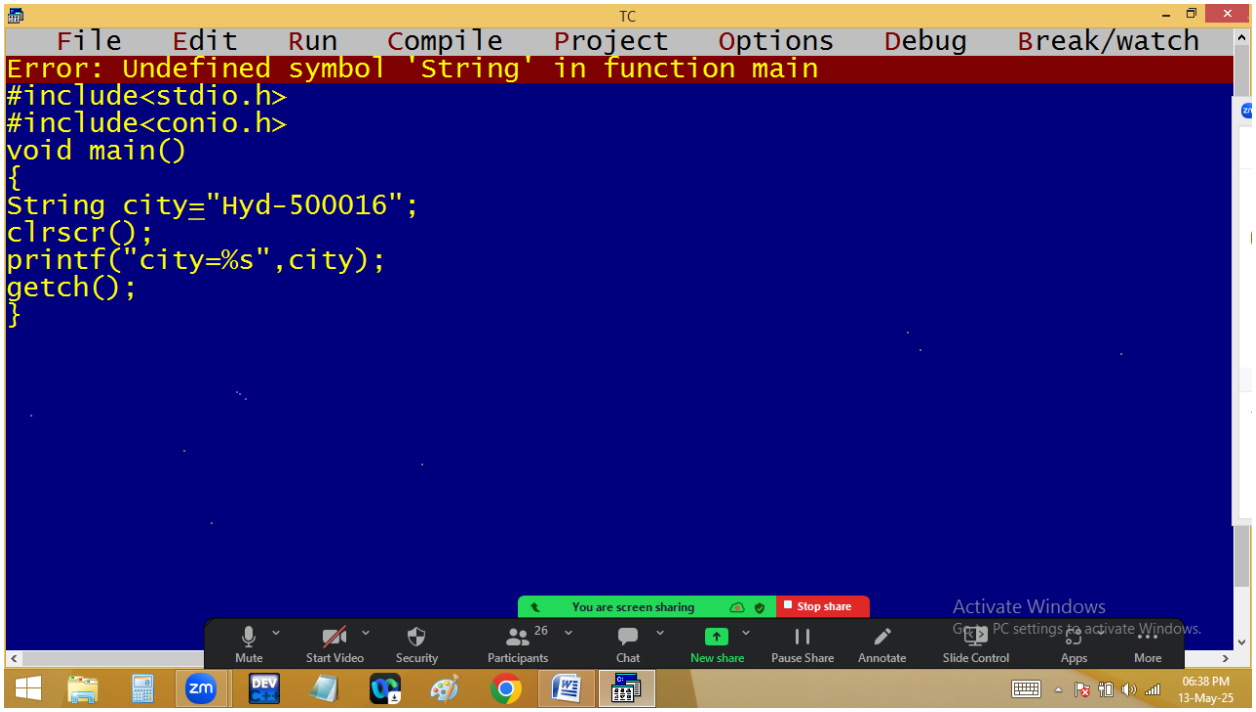
String – group of characters is called string.
to store alpha-numeric we are using string. i.e. alphabets, numbers and special characters

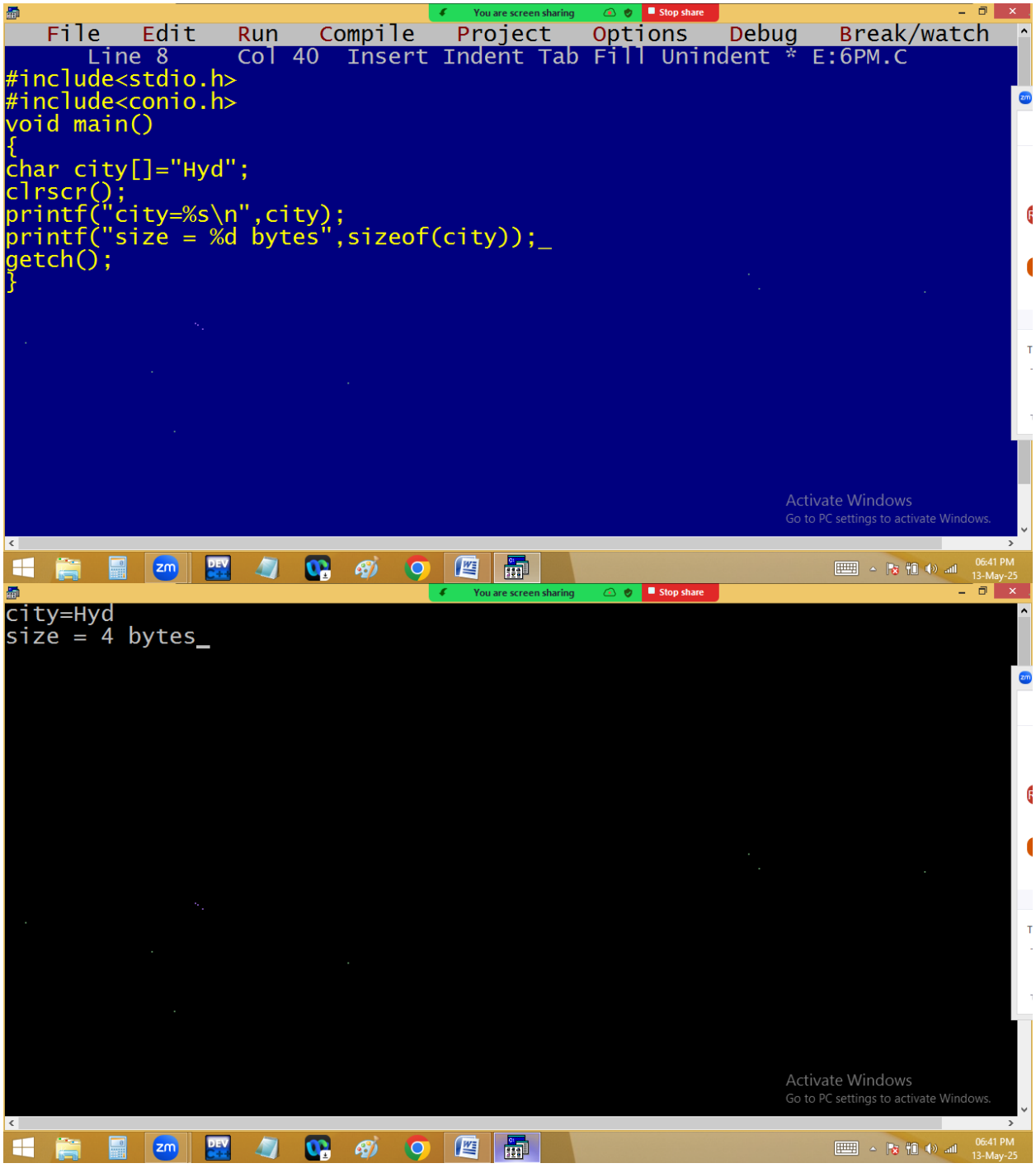
Eg: char city[20]="Hyd-500016"; ➔ %s ➔ 20 bytes

Char array - string

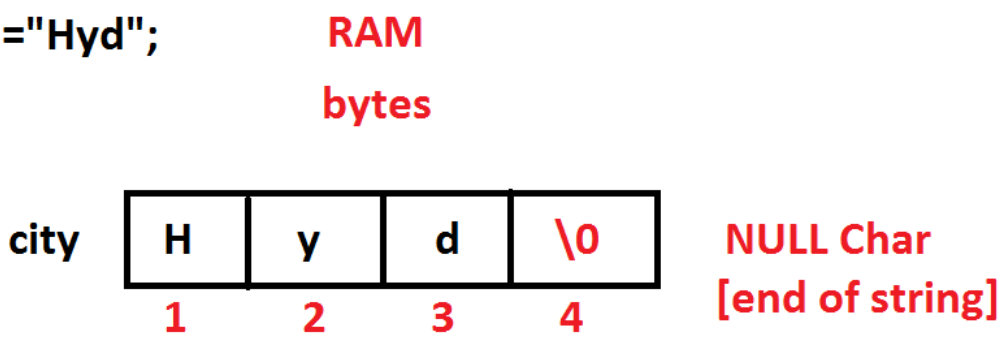
String name = "Hyd-500016";


TC
File Edit Run Compile Project Options Debug Break/watch
Line 7 Col 24 Insert Indent Tab Fill Unindent * E:6PM.C
#include<stdio.h>
#include<conio.h>
void main()
{
char city[20]="Hyd-500016";
clrscr();
printf("city=%s",city);
getch();
}
Activate Windows
Go to PC settings to activate Windows.
You are screen sharing
Stop share
06:37 PM
13-May-25
TC
city=Hyd-500016_
You are screen sharing
Stop share
Activate Windows
Mute
Start Video
Security
Participants
26
Chat
New share
Pause Share
Annotate
Slide Control
Apps
More
06:37 PM
13-May-25


TC
File Edit Run Compile Project Options Debug Break/watch
Error: Undefined symbol 'String' in function main
#include<stdio.h>
#include<conio.h>
void main()
{
String city="Hyd-500016";
clrscr();
printf("city=%s",city);
getch();
}
You are screen sharing
Stop share
Mute
Start Video
Security
Participants
26
Chat
New share
Pause Share
Annotate
Slide Control
Apps
More
Activate Windows
06:38 PM
13-May-25


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Stop share
File Edit Run Compile Project Options Debug Break/watch
Line 8 Col 40 Insert Indent Tab Fill Unindent * E:6PM.C
#include<stdio.h>
#include<conio.h>
void main()
{
char city[]="Hyd";
clrscr();
printf("city=%s\n",city);
printf("size = %d bytes",sizeof(city));
getch();
}
Activate Windows
Go to PC settings to activate Windows.
06:41 PM
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You are screen sharing
Stop share
city=Hyd
size = 4 bytes
Activate Windows
Go to PC settings to activate Windows.
06:41 PM
13-May-25

char city[ ]="Hyd";



Void - EMPTY

Derived data types: They created from primitive data types.

1. Array
   Eg:
   int a[20]; ← int array type variable – non primitive
   char city[20]; ← char array type variable – non primitive
2. Pointer - int * p; → int pointer variable
3. Function - int sum( ); → int function

User defined data types: Created by the user by using of both primitive and derived data types.

1. Structure
2. Union
3. enumeration

DATA TYPES

Data type determines the type of value we are going to store in our computer. To store anything in our computer, we should have to allocate the memory. This memory allocation is depended on the data type we are using.

Data type determines the properties such as

1. No of bytes
2. Range
3. Type of value

In C language we are having 3 **basic** data types

1. **Int – To store non-decimal numbers**
2. **Float – To store decimal numbers**
3. **Char – To stores alphabets, numbers and special char**

Total data types are divided into 3 types.

1. **Primitive data types**
2. **Derived data types**
3. **User defined data types**

PRIMITIVE DATA TYPES:

These are the regular data types we are using in our c programs.

Data type	Bytes In 16 bit compiler	Conversion Character / format specifier	Storage Range
int / signed int / short int	2	%d	-32768 to +32767
unsigned int	2	%u	0 to 65535
long int	4	%ld	-2147483648 to 2147483647
unsigned long int	4	%lu	0 to 4294967295
Float	4	%f	$3.4 * 10^{-38}$ to $3.4 * 10^{+38}$
double	8	%lf	$1.7 * 10^{-308}$ to $1.7 * 10^{+308}$
long double	10	%Lf	$3.4 * 10^{-4932}$ to $1.1*10^{+4932}$
Char	1	%c	1 character Signed char [-128 to +127] Unsigned char [0 to 255]
char[10] (STRING)	10	%s	9 char + 1 null char
void [empty data type]			nothing

Note: In 32 bit compilers like dev c++ / vs code / online compilers / java / .net / python int takes 4 bytes.

DERIVED DATA TYPES:

They are derived from primitive data types.

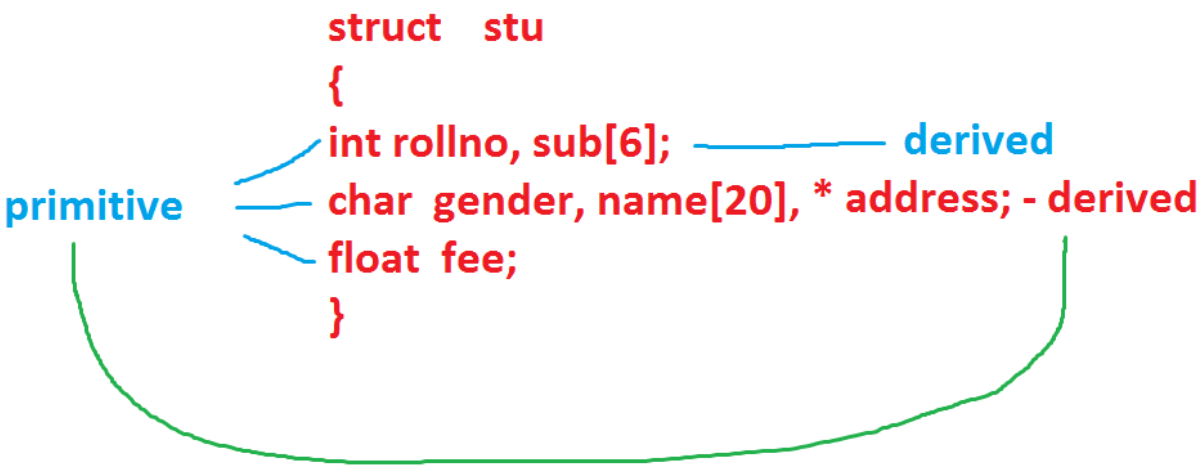
1. **Array [non-primitive]**
2. **Pointer**
3. **Function**

USER DEFINED DATA TYPES:

These are the data types created by the user.

1. Structure

   Eg:



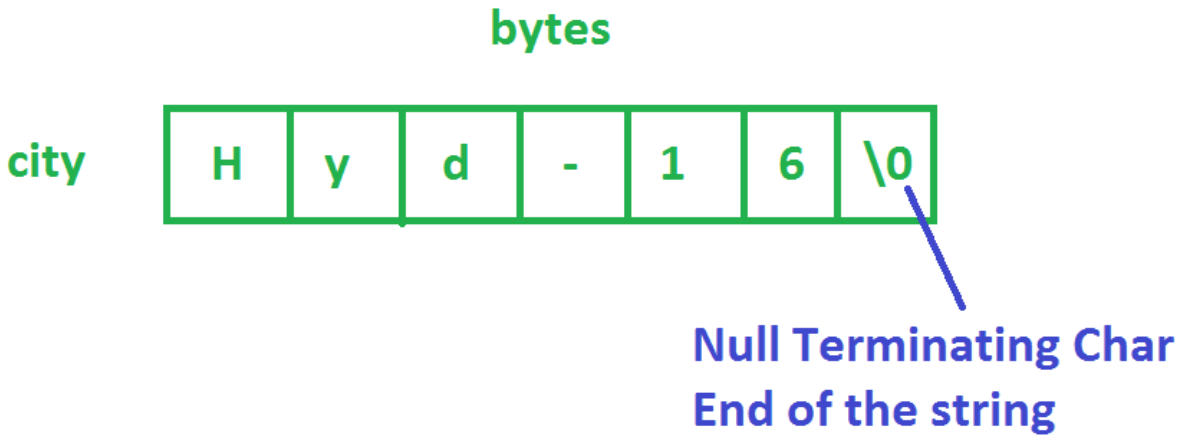
2. union
3. enum

String:

1. A group of characters is called string.
2. It is alpha-numeric.

3. It is a char array.

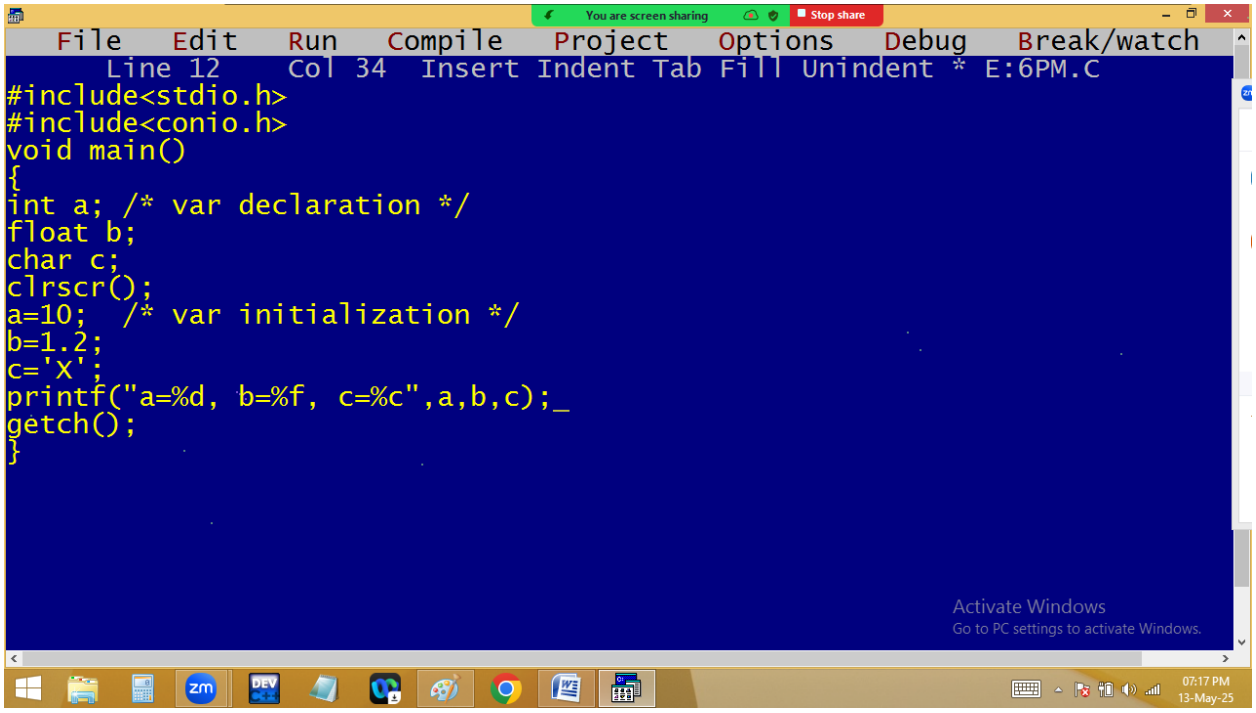
   Eg: char city[7]=”**Hyd-16**”;

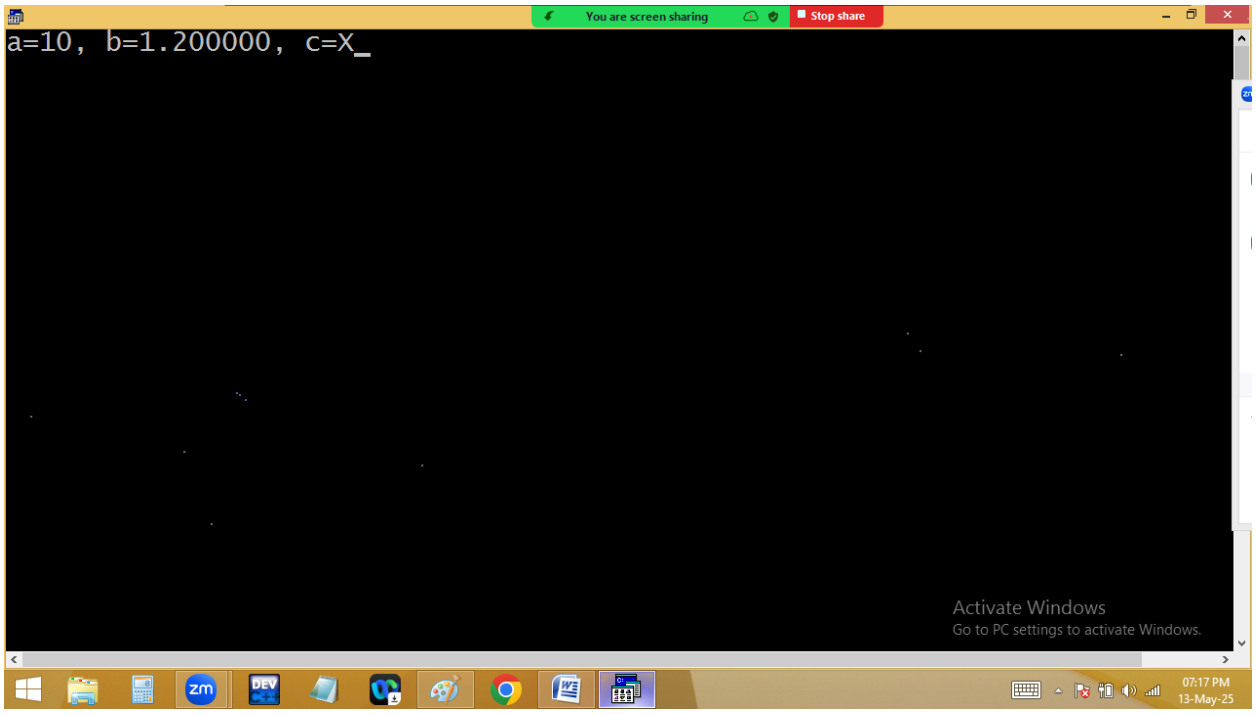


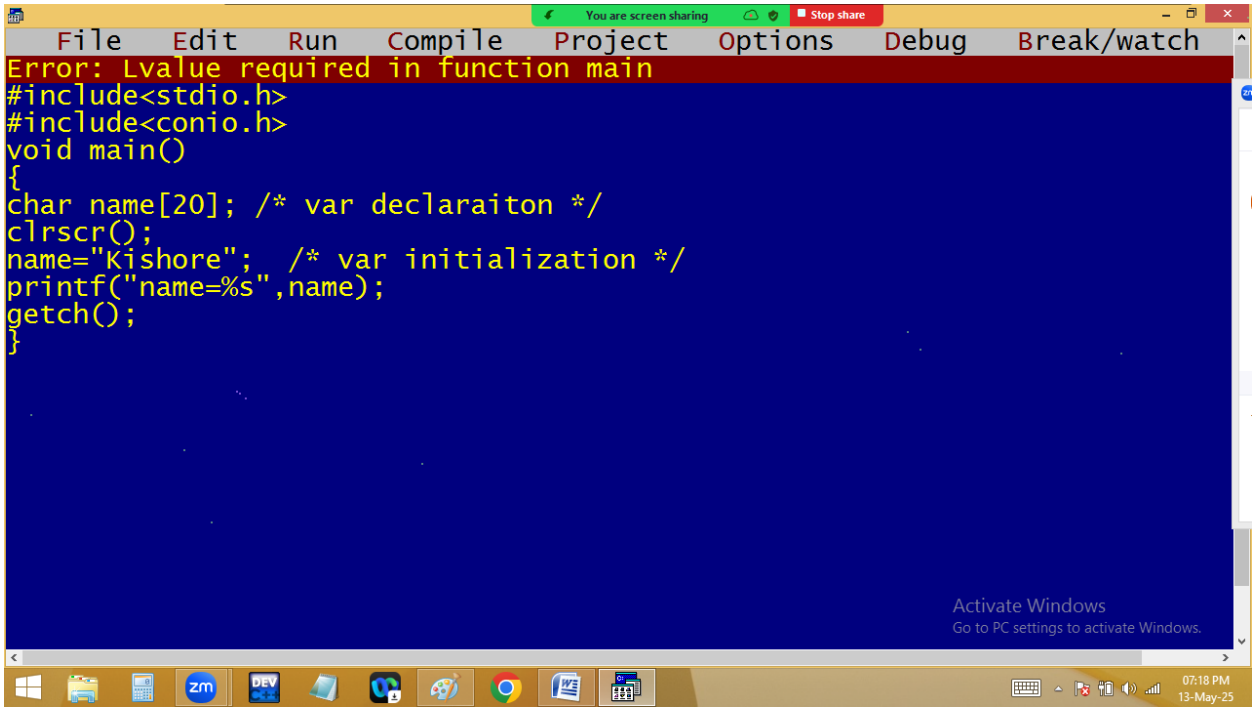
Note:

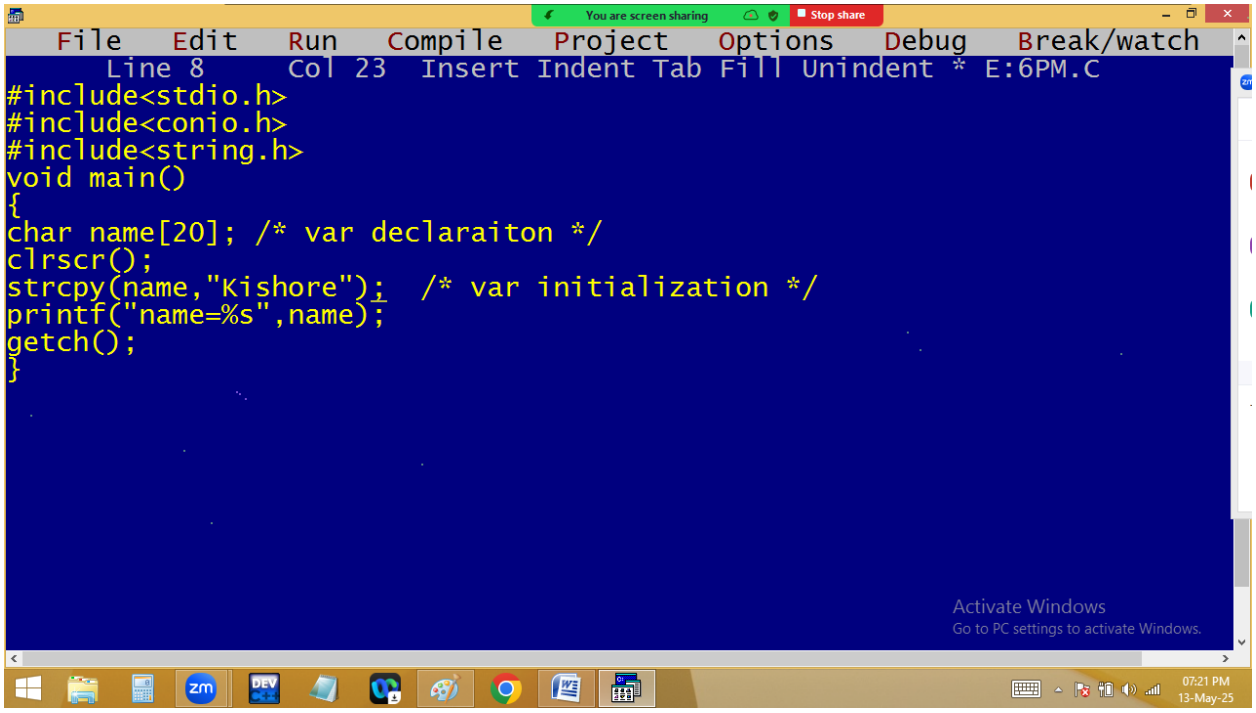
We can’t copy a string using = operator in C & C++.

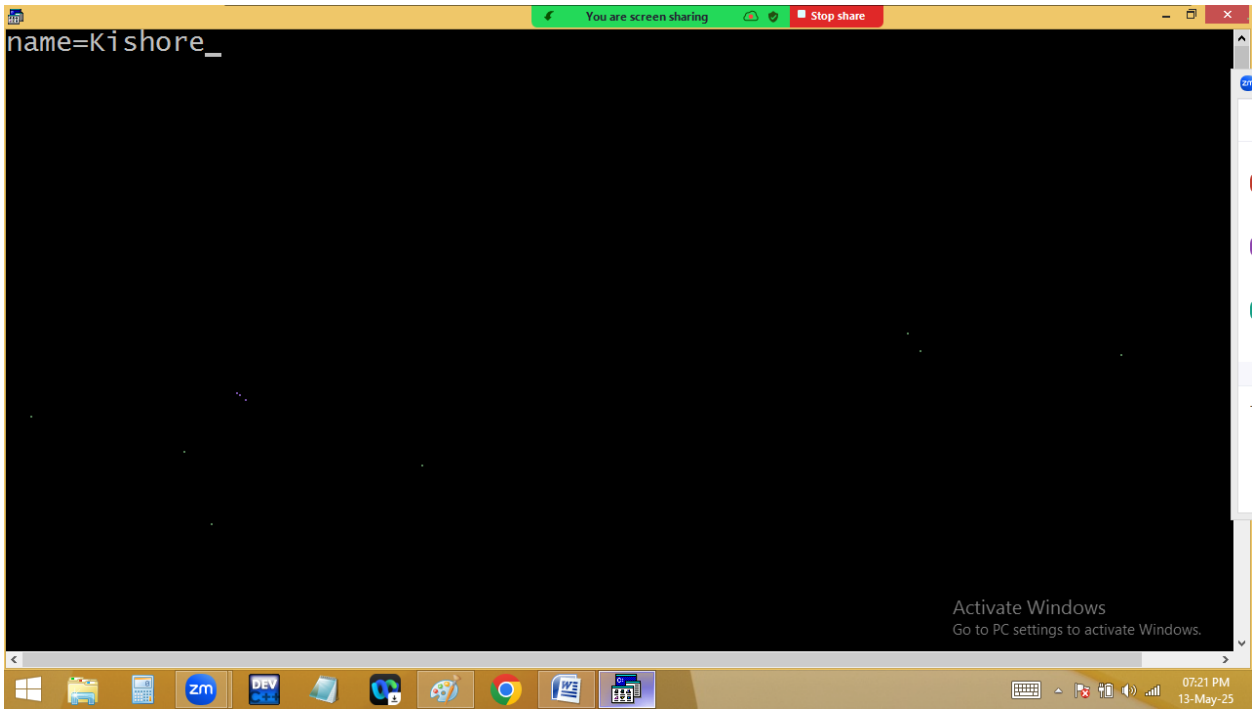
For this we have to use **strcpy( )** available in <string.h> or we have to copy character by character.


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File Edit Run Compile Project Options Debug Break/watch
Line 12 Col 34 Insert Indent Tab Fill Unindent * E:6PM.C
#include<stdio.h>
#include<conio.h>
void main()
{
int a; /* var declaration */
float b;
char c;
clrscr();
a=10; /* var initialization */
b=1.2;
c='X';
printf("a=%d, b=%f, c=%c",a,b,c);_
getch();
}
Activate Windows
Go to PC settings to activate Windows.
07:17 PM
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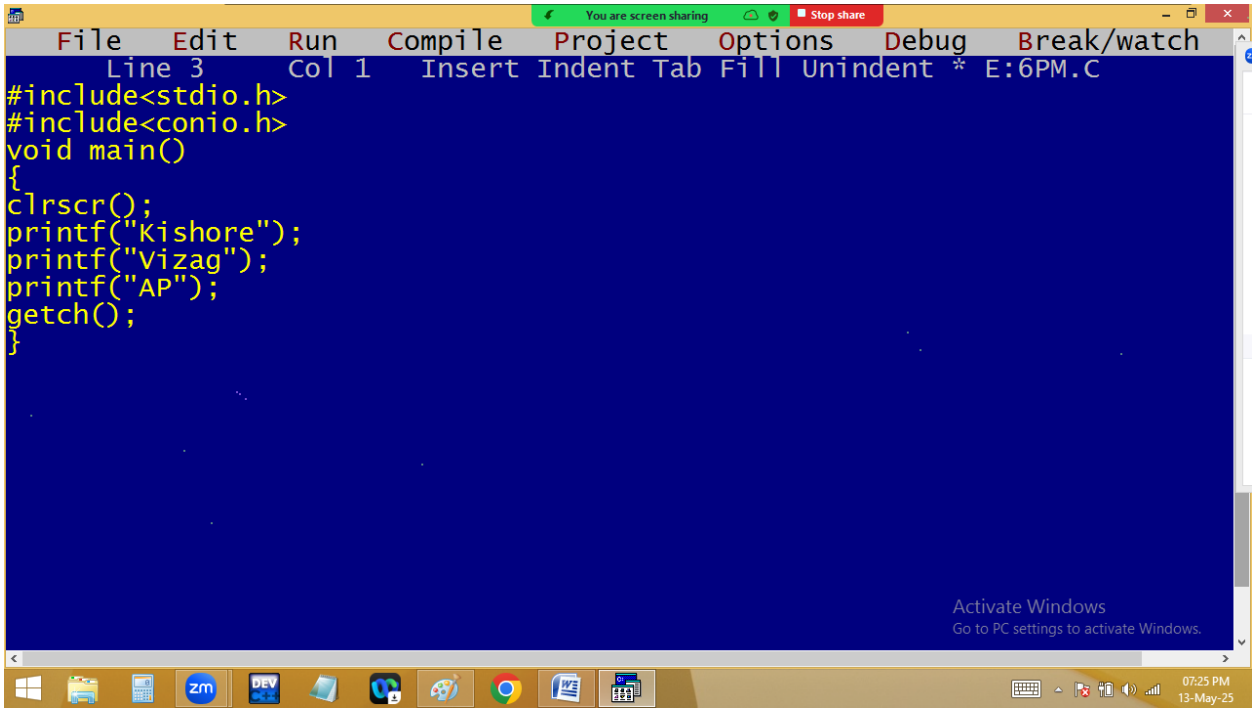

You are screen sharing
Stop share
a=10, b=1.200000, c=X_
Activate Windows
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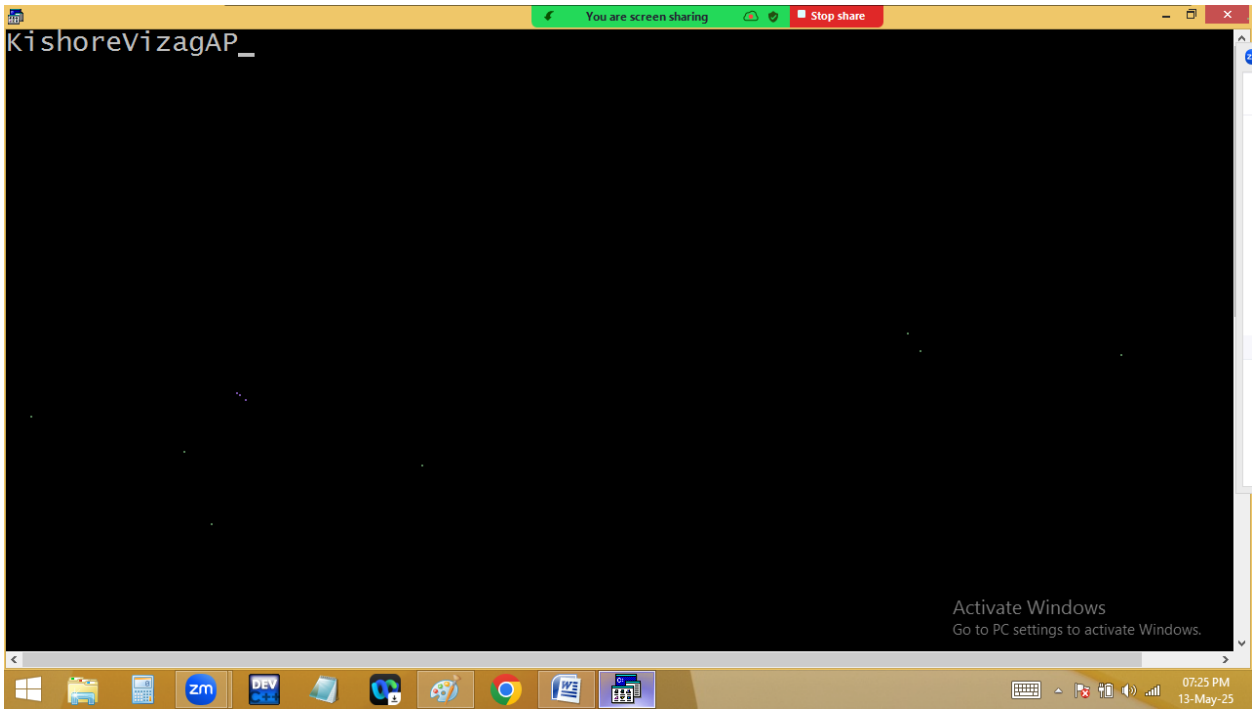

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File Edit Run Compile Project Options Debug Break/watch
Error: Lvalue required in function main
#include<stdio.h>
#include<conio.h>
void main()
{
char name[20]; /* var declaraiton */
clrscr();
name="Kishore"; /* var initialization */
printf("name=%s",name);
getch();
}
Activate Windows
Go to PC settings to activate Windows.
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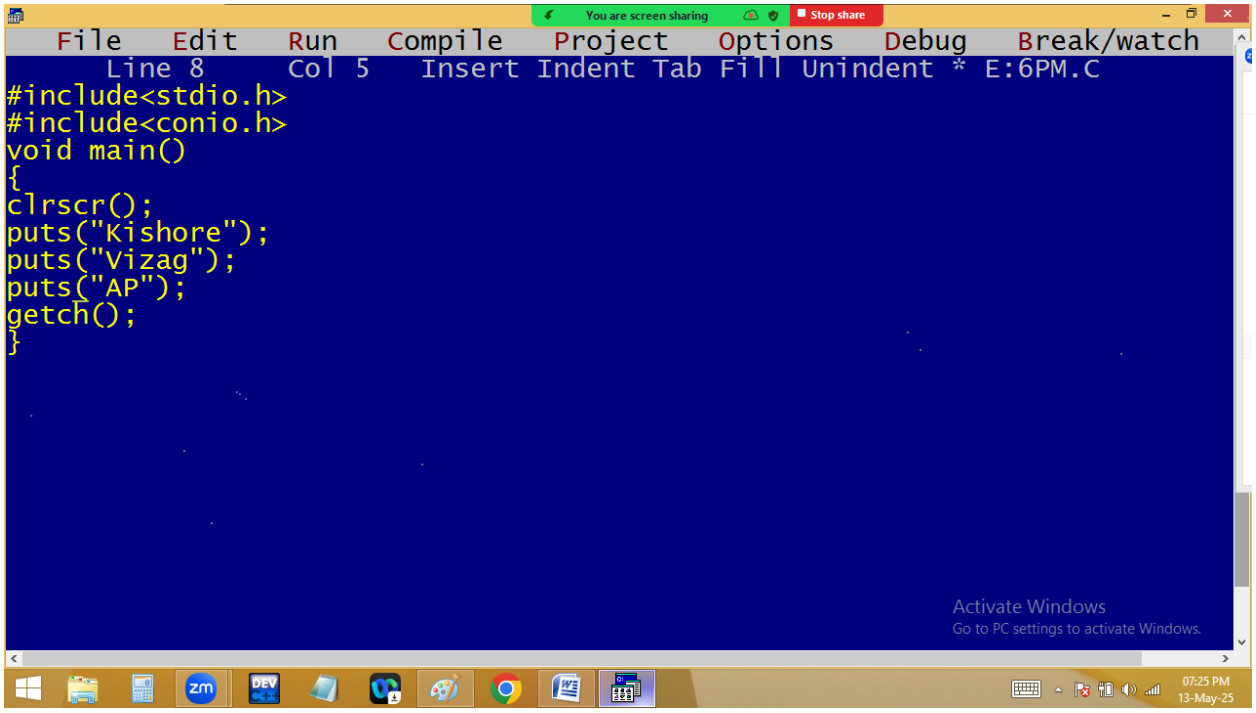


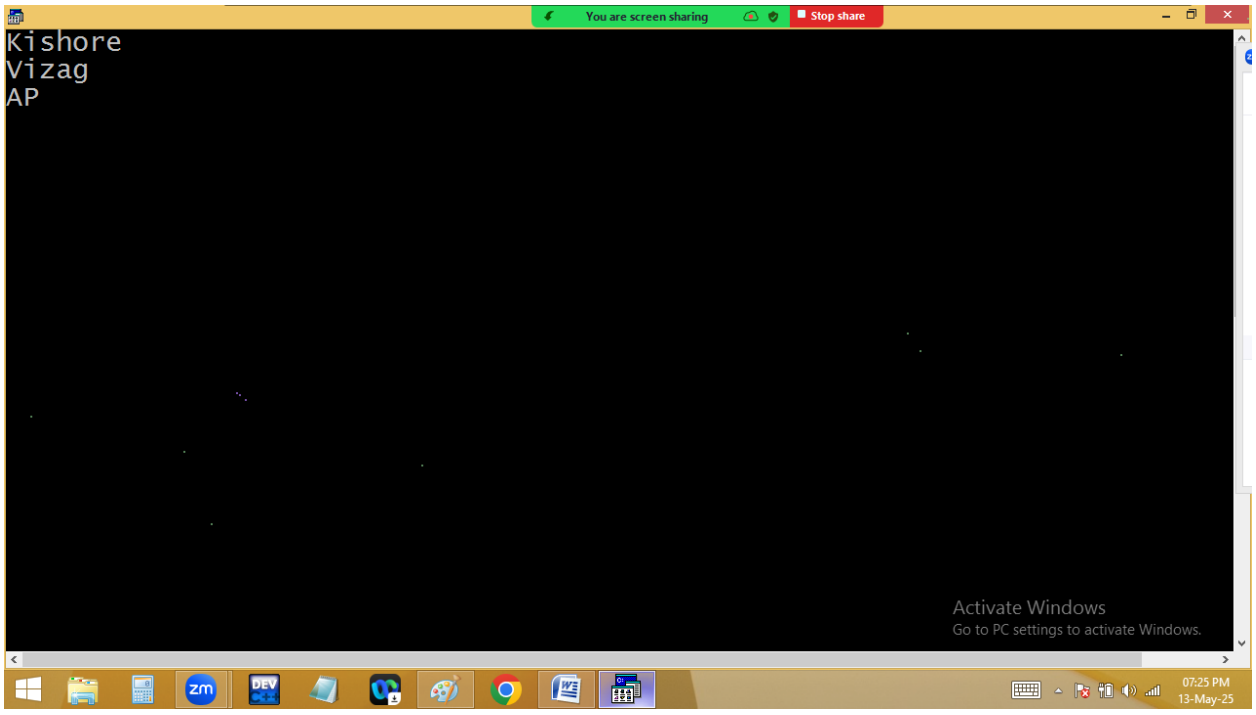


Note: we can't compare two strings using == operator. We have to strcmp() available in string.h.


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File Edit Run Compile Project Options Debug Break/watch
Line 3 Col 1 Insert Indent Tab Fill Unindent * E:6PM.C
#include<stdio.h>
#include<conio.h>
void main()
{
clrscr();
printf("Kishore");
printf("Vizag");
printf("AP");
getch();
}
Activate Windows
Go to PC settings to activate Windows.
07:25 PM
13-May-25

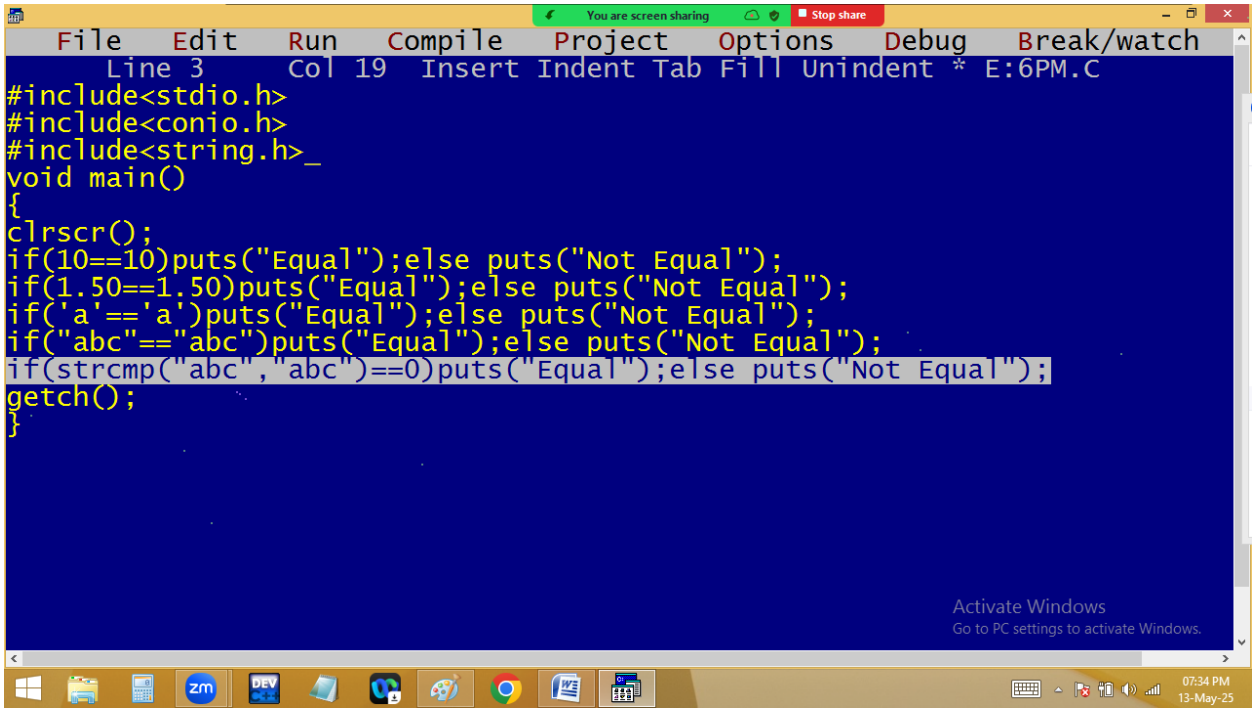

You are screen sharing
Stop share
KishoreVizagAP_
Activate Windows
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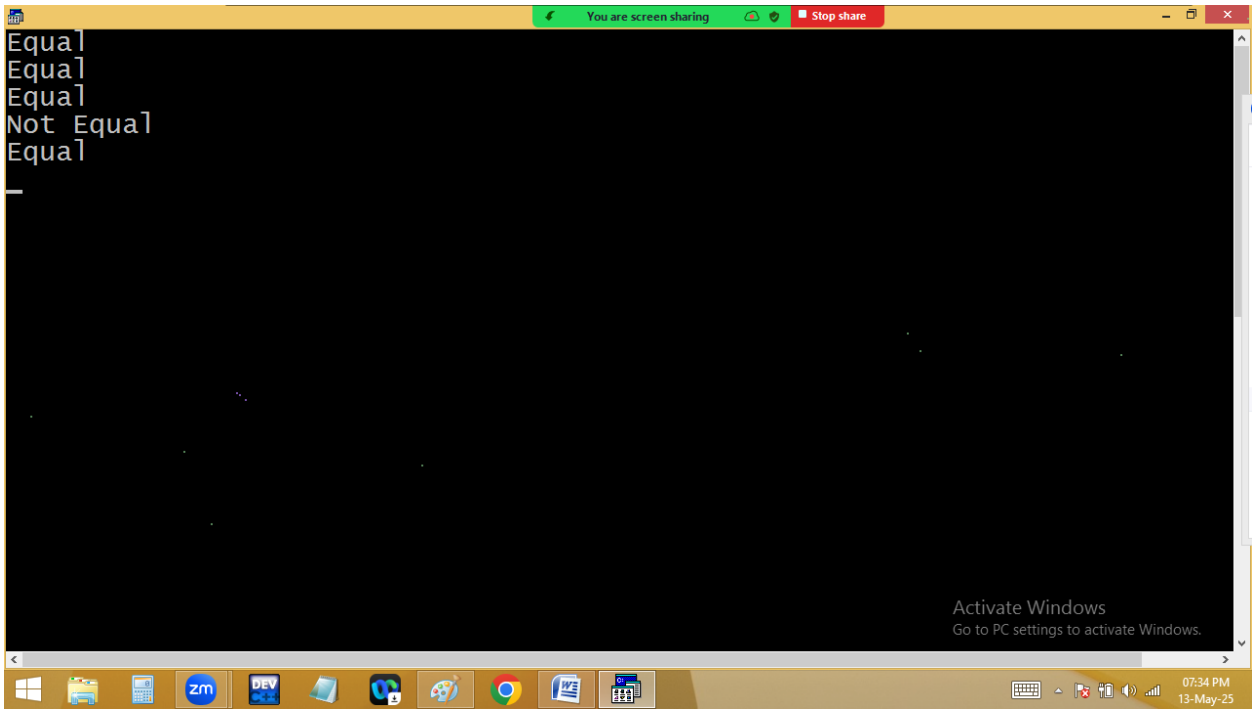

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Line 8 Col 5 Insert Indent Tab Fill Unindent * E:6PM.C
#include<stdio.h>
#include<conio.h>
void main()
{
clrscr();
puts("Kishore");
puts("Vizag");
puts("AP");
getch();
}
Activate Windows
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You are screen sharing
Stop share
Kishore
Vizag
AP
Activate Windows
Go to PC settings to activate Windows.
07:25 PM
13-May-25

$$\begin{array}{r} 4 \\ -\,4 \\ \hline 0 \end{array} \qquad \begin{array}{r} 4 \\ -\,3 \\ \hline 1 \end{array} \qquad \begin{array}{r} 3 \\ -\,4 \\ \hline -1 \end{array}$$

$$\begin{array}{r} 97\,b\,c \\ -\,97\,b\,c \\ \hline \end{array}$$


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File Edit Run Compile Project Options Debug Break/watch
Line 3 Col 19 Insert Indent Tab Fill Unindent * E:6PM.C
#include<stdio.h>
#include<conio.h>
#include<string.h>_
void main()
{
clrscr();
if(10==10)puts("Equal");else puts("Not Equal");
if(1.50==1.50)puts("Equal");else puts("Not Equal");
if('a'=='a')puts("Equal");else puts("Not Equal");
if("abc"=="abc")puts("Equal");else puts("Not Equal");
if(strcmp("abc","abc")==0)puts("Equal");else puts("Not Equal");
getch();
}
Activate Windows
Go to PC settings to activate Windows.
07:34 PM
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You are screen sharing
Stop share
Equal
Equal
Equal
Not Equal
Equal
Activate Windows
Go to PC settings to activate Windows.
07:34 PM
13-May-25